DSS5202 Sustainable Systems Analysis
Assignment 4 Solutions

Question 1.

The LP model:

Sets

$$P = \{\text{General},\ \text{Fresh-Food},\ \text{Fragile}\} \quad \text{// set of parcel categories,}$$
$$M = \{\text{RPB},\ \text{ASP},\ \text{LCM}\} \quad \text{// set of packaging methods.}$$

Parameters

c_{ij} = cost for processing category i parcels using packaging method j, $\quad i \in P,\ j \in M,$
d_i = volume of category i parcels processed annually, $\quad i \in P,$
e_i = current packaging material consumption of category i parcels, $\quad i \in P,$
r_j = material reduction rate of packaging method j, $\quad j \in M,$
t = target fraction of total material reduction.

Data

$$\mathbf{c} = \begin{bmatrix} 2.2 & 1.5 & 0.9 \\ 3.1 & 1.9 & 1.1 \\ 2.8 & 2.1 & 1.3 \end{bmatrix} \qquad \mathbf{d} = \begin{bmatrix} 160 \\ 90 \\ 120 \end{bmatrix}$$

$$\mathbf{e} = \begin{bmatrix} 0.36 \\ 0.64 \\ 0.52 \end{bmatrix} \qquad \mathbf{r} = \begin{bmatrix} 0.80 \\ 0.65 \\ 0.45 \end{bmatrix}$$

$t = 0.75$

Decision Variables

x_{ij} = annual volume of category i parcels that use packaging method j, $\quad i \in P,\ j \in M.$

Model

$$\text{Min} \quad \sum_{i \in P} \sum_{j \in M} c_{ij} x_{ij} \qquad (1)$$

$$\text{s.t.} \quad \sum_{j \in M} x_{ij} = d_i \qquad \forall\, i \in P \qquad (2)$$

$$\sum_{i \in P} \sum_{j \in M} r_j e_i x_{ij} \geq t \sum_{i \in P} e_i d_i \qquad (3)$$

$$x_{ij} \geq 0 \qquad \forall\, i \in P,\ \forall\, j \in M \qquad (4)$$

- Objective function (1) minimizes the total annual cost.
- Constraint set (2) ensures that all parcels are processed using a packaging method.
- Constraint (3) ensures that the minimum required material reduction fraction is achieved.
- Constraint set (4) defines the domain of the decision variables.

Question 2.

Solutions obtained using the Gurobi solver:

Total annual cost: $852,000

The number of parcels processed annually using each method:

Parcel category	Reusable Packaging Boxes (RPB)	Automated Size-Matched Packaging (ASP)	Lightweight Cushioning Materials (LCM)
General parcels	0	160,000	0
Fresh-food parcels	87,500	2,500	0
Fragile parcels	120,000	0	0

Current total annual packaging material (kg): 177,600

Annual packaging material reduced (kg): 133,200

Total percentage reduction achieved (%): **75%**

Question 3.

The LP model in Question 1 is extended to an MIP as follows:

Additional parameter

$\text{big}M = \text{Big number}$

Additional Decision Variables

$$y_{ij} = \begin{cases} 1 & \text{if category } i \text{ parcels uses packaging method } j \\ 0 & \text{otherwise} \end{cases} \qquad \forall\, i \in P,\ \forall\, j \in M.$$

Model

Additional constraints (5) to (7) are added:

$$\text{Min} \quad \sum_{i \in P} \sum_{j \in M} c_{ij} x_{ij} \qquad (1)$$

$$\text{s.t.} \quad \sum_{j \in M} x_{ij} = d_i \qquad \forall\, i \in P \qquad (2)$$

$$\sum_{i \in P} \sum_{j \in M} r_j e_i x_{ij} \geq t \sum_{i \in P} e_i d_i \qquad (3)$$

$$x_{ij} \geq 0 \qquad \forall\, i \in P,\ \forall\, j \in M \qquad (4)$$

$$\sum_{j \in M} y_{ij} \leq 1 \qquad \forall\, i \in P \qquad \textbf{(5)}$$

$$x_{ij} \leq \text{big}M\, y_{ij} \qquad \forall\, i \in P,\ \forall\, j \in M \qquad \textbf{(6)}$$

$$y_{ij} \in \{0, 1\} \qquad \forall\, i \in P,\ \forall\, j \in M \qquad \textbf{(7)}$$

- Constraint sets (5) and (6) ensure that only 1 packaging method is assigned for each parcel category.
- Constraint set (7) defines the domain of the additional decision variables.

Question 4.

Using the Gurobi solver:

Total annual cost: $855,000

The number of parcels processed annually using each method:

Parcel category	Reusable Packaging Boxes (RPB)	Automated Size-Matched Packaging (ASP)	Lightweight Cushioning Materials (LCM)
General parcels	0	160,000	0
Fresh-food parcels	90,000	0	0
Fragile parcels	120,000	0	0

Current total annual packaging material (kg): 177,600

Annual packaging material reduced (kg): 133,440

Total percentage reduction achieved (%): **75.14%**